

NFL Predictive Wins Model

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Abstract:

Background

Sports analytics research increasingly focuses on predicting team performance, yet most existing NFL predictive win models rely on cross sectional or binary outcome frameworks. Fewer studies model longitudinal, count based outcomes.

Study Design

Using NFL data from 2003–2023, a predictive modeling framework was developed. Predictors were selected based on data availability, subject matter expertise, and prior literature.

Methods

A generalized estimating equation model (GEE) with Poisson, Quasi Poisson, and Negative Binomial specifications were fit. Model selection was conducted using QIC based backward selection within model families, and predictive performance was evaluated using Root Mean Square Error (RMSE).

Results

The Quasi Poisson GEE model demonstrated the best balance of robustness and predictive accuracy, achieving an RMSE of 1.59 on the 2024 season.

Conclusion

This study demonstrates that count based longitudinal models using GEE provide an effective framework for predicting NFL team wins. The proposed approach extends prior work and offers a practical, interpretable alternative to purely cross sectional or machine learning based models.

Introduction:

The NFL is the most popular professional team sports league in the US. Similar to other major American sports leagues, such as Major League Baseball and the National Basketball Association, the NFL has experienced a substantial growth in the use of data driven analytics over recent decades to inform team decisions. In any league, identifying a reliable “winning formula” is vital, making the ability to accurately predict team wins both practically and analytically valuable.

The primary objective of this study was to develop a predictive model for forecasting NFL regular season team wins based on a set of key predictors. Several studies have previously explored predictive modeling of NFL outcomes. *Advancing NFL Win Prediction: From Pythagorean Formulas to Machine Learning Algorithms* (Weirich et al., 2025) examined the evolution of NFL win prediction methods, utilizing traditional statistical approaches such as modern machine learning techniques, including decision trees and logistic regression. Similarly, *A Predictive Analytics Model for Forecasting Outcomes in the National Football League Games Using Decision Tree and Logistic Regression* (Gifford & Bayrak, 2023) applied decision tree and binary logistic regression models to predict individual game outcomes using team performance metrics.

While informative, both studies focused on binary win outcomes. Moreover, none of these prior works explicitly account for the longitudinal nature of team performance by modeling season level win totals as repeated count outcomes over time. This study addresses this by developing a longitudinal, population averaged modeling framework that captures within team correlation across seasons, to predict NFL team success.

Methods:

The NFL consists of 32 teams. This study analyzed data from the 2003–2023 regular seasons, covering a total of 21 seasons. Each NFL team was treated as a statistical cluster, with one observation contributed per team per season.

Variables and Model Specification

The outcome of interest was the total number of regular season wins per team per season, modeled as a count variable. The predictor variables selected were offensive ranking (offense), defensive ranking (defense), previous year playoff participation (playoff), average roster age (age), and head coach tenure (coaching).

A population averaged marginal count modeling framework was adopted using generalized estimating equations (GEE). A Poisson, Quasi Poisson, and Negative Binomial distribution were initially specified to account for the count nature of the outcome and within team correlation across seasons.

The initial model assessed was the following:

$$\log(\mu_{it}) = \log(G_{it}) + \beta_0 + \beta_1 \text{Offense}_{it} + \beta_2 \text{Defense}_{it} + \beta_3 \text{Playoffs}_{i,t-1} + \beta_4 \text{Age}_{it} + \beta_5 \text{Coaching}_{it} + \beta_6 (\text{Offense}_{it} \times \text{Playoffs}_{i,t-1}) + \beta_7 (\text{Age}_{it} \times \text{Defense}_{it}) + \beta_8 (\text{Age}_{it} \times \text{Playoffs}_{i,t-1}) + \beta_9 (\text{Age}_{it} \times \text{Coaching}_{it})$$

where Y_{it} = number of wins for team i in season t , G_{it} = games played (offset) and $Y_{it} \sim \text{Count}(\mu_{it})$

Model Selection and Comparison

The Quasi likelihood under the Independence model Criterion (QIC) was used for model selection within the GEE framework. QIC values were used to compare models within each distributional family (Poisson, Quasi Poisson, and Negative Binomial), and backward selection was applied to identify the best performing model for each distribution. Because QIC is not directly comparable across different quasi likelihood families, cross model comparisons were conducted using predictive performance metrics.

Predictive Evaluation

Final models for each respective distributional family were evaluated using Root Mean Square Error (RMSE) calculated on both the training data and an independent testing dataset (2024 season). RMSE was used to assess and compare the out of sample predictive accuracy of competing model specifications.

Results:

Table 1. Descriptive Characteristics of the NFL Study Sample, Stratified by Season (Team Level Aggregates)

Seasons	Offense; Frequency				Defense; Frequency				Playoffs; Frequency	Age; Mean (SD)	Coaching; Mean (SD)
	1	2	3	4	1	2	3	4			
2003	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.62)	1.84 (0.37)
2004	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.58)	2.41 (0.84)
2005	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.58)	3.16 (1.08)
2006	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	27.0 (0.51)	3.06 (1.70)
2007	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	27.0 (0.49)	3.34 (2.01)
2008	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	27.0 (0.54)	3.94 (2.20)
2009	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.52)	3.44 (2.50)
2010	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.8 (0.57)	4.19 (2.66)
2011	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.7 (0.49)	4.11 (2.87)
2012	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.6 (0.46)	4.33 (3.16)
2013	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.4 (0.51)	4.09 (3.17)
2014	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.4 (0.42)	4.27 (3.48)
2015	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.5 (0.49)	4.38 (3.85)
2016	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.1 (0.41)	4.59 (3.93)
2017	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.0 (0.46)	5.03 (4.28)
2018	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.0 (0.41)	5.23 (4.75)
2019	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.0 (0.42)	4.83 (4.52)
2020	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.3 (0.47)	4.97 (4.63)
2021	25%	25%	25%	25%	25%	25%	25%	25%	43.8%	26.3 (0.54)	5.00 (5.04)
2022	25%	25%	25%	25%	25%	25%	25%	25%	43.8%	26.4 (0.45)	4.59 (5.04)
2023	25%	25%	25%	25%	25%	25%	25%	25%	43.8%	26.5 (0.43)	5.14 (5.27)

SD = Standard Deviation

All statistics calculated on n = 32 observations (32 NFL teams)

Offensive and defensive rank categories each contained 25% of teams per season as team rankings were defined using seasonal quartiles. Playoff participation increased to 43.8% beginning in 2021, reflecting the NFL's expansion from 12 to 14 playoff teams. The average roster age remained relatively stable across seasons. In contrast, average coaching tenure increased over time.

The final Poisson, Quasi Poisson and Negative Binomial GEE models, retained using backward selection based on the QIC included all main effects. For the Poisson model, all interaction terms were kept. The Negative Binomial model retained interactions between age and coaching and offense and playoffs, whereas the Quasi Poisson model retained only age and coaching.

The predictive performance of each model was evaluated using RMSE on the training data. The Poisson GEE and Quasi Poisson GEE models demonstrated similar predictive accuracy (RMSE = 1.62 and 1.63, respectively), whereas the Negative Binomial GEE model exhibited substantially poorer performance (RMSE = 4.15).

RMSE was subsequently evaluated using out of sample testing data from the 2024 season. The Quasi Poisson GEE model achieved the lowest RMSE (1.59), outperforming the Poisson GEE model (RMSE = 1.65). Consequently, the Quasi Poisson GEE model was selected as the final predictive model.

The final Quasi Poisson GEE model provides some insights into the drivers of NFL team wins.

Table 2. Estimated Regression Coefficients from the Final Quasi Poisson GEE Model for NFL Team Wins

Term	Estimate	Std. Error	Significance
Intercept	-1.8592	0.5953	**
Offensive Rating (Category 2 vs 1)	-0.1616	0.0186	***
Offensive Rating (Category 3 vs 1)	-0.3428	0.0225	***
Offensive Rating (Category 4 vs 1)	-0.6536	0.0296	***
Defensive Rating (Category 2 vs 1)	-0.1424	0.0187	***
Defensive Rating (Category 3 vs 1)	-0.2544	0.0208	***
Defensive Rating (Category 4 vs 1)	-0.5408	0.0280	***
Made Playoffs Previous Season (Yes vs No)	0.0203	0.0167	—
Average Team Age (years)	0.0612	0.0224	**
Coaching Tenure (years)	0.2923	0.1091	**
Average Age × Coaching Tenure	-0.0111	0.0041	**

Significance: *** p < 0.001 ** p < 0.01 * p < 0.05

As offense and defense worsens, expected win totals decrease monotonically. Prior year playoff participation does not have a statistically significant effect on current season win totals. Age and coaching are positively associated with expected wins, implying that older teams and experienced coaching, lead to more wins. The significant interaction between age and coaching tenure indicates that older teams appear to benefit more from experience when led by new coaches, whereas for long tenured coaches, the marginal advantage of an older roster is reduced.

This study developed a longitudinal, population averaged model to predict NFL regular season wins. Among the three competing models, the Quasi Poisson GEE model demonstrated the strongest overall predictive performance. When evaluated on the 2024 season, it achieved an RMSE of 1.59, indicating that predicted win totals were, on average, within approximately 1 to 2 wins of the observed outcomes. This level of error suggests reasonable and practically meaningful predictive accuracy, particularly given the inherent uncertainty in NFL season outcomes.

Limitations:

This study relies on aggregated team level predictors such as offensive and defensive rankings, average roster age, prior year playoff appearance and coaching tenure. As a result, potentially important factors including injuries, quarterback changes, schedule strength, in season roster adjustments were not captured in the model. Moreover, psychological factors are widely recognized as important influences on individual and team performance in the NFL. Factors such as coaching philosophy, player IQ, leadership, team chemistry, and in game decision making are difficult to quantify and were therefore not included in the model, potentially limiting its ability to fully capture performance variability. Additionally, predictive performance was evaluated using a single out of sample season (2024), which limits conclusions about the model's stability and generalizability across multiple future seasons.

References:

Weirich C, Kim JW, Yoon Y, Jeong S. Advancing NFL win prediction: from Pythagorean formulas to machine learning algorithms. *Front Sports Act Living*. 2025 Sep 12;7:1638446. doi: 10.3389/fspor.2025.1638446. PMID: 41018816; PMCID: PMC12463883.

Matt Gifford, Tuncay Bayrak, A predictive analytics model for forecasting outcomes in the National Football League games using decision tree and logistic regression, *Decision Analytics Journal*, Volume 8, 2023,100296, ISSN 2772-6622, <https://doi.org/10.1016/j.dajour.2023.100296>.